

Paper Agent: A Traceable Multi-Agent System for Verified Scholarly Paper Discovery

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Abstract

Business-school literature review requires more than typically plausible recommendations. Researchers must verify DOI metadata, enforce journal-selection policies, inspect open-access status, and retain reproducible evidence explaining why each paper was included or excluded. A single large-language-model (LLM) response can produce useful suggestions, but it can also return unverifiable metadata and opaque recommendations. We present *Paper Agent*, a traceable multi-agent system for verified scholarly paper discovery. The system decomposes literature review into a 12-stage tool-using pipeline covering planning, journal selection, retrieval, journal filtering, metadata verification, open-access checking, storage, evaluation preparation, relevance scoring, ranking, critic review, and report delivery. The deployed workflow uses Web of Science as its primary configured scholarly search provider and OpenAlex as a fallback, Crossref for DOI and metadata verification, Unpaywall for open-access status, Cloudflare Workers for execution, and Cloudflare D1 for persistent traces.

We evaluate the system with *Paper-Agent-Bench*. Quantitative comparison is restricted to a verified controlled T001–T003 layer containing nine comparison rows. On this layer, the Single LLM baseline has higher Precision@5 and NDCG@5, while Paper Agent preserves DOI accuracy, strict top-journal precision, traceability, and explicit policy enforcement. We additionally report artifact-only execution evidence: a gated T004–T006 dry-run produced three job rows and 50 result rows, and a staged T007–T012 attempt produced 87 result rows for T008–T012 while T007 timed out after approximately 250 seconds. Later batches were not started. These expansion artifacts are not benchmark validation. Full T004–T020 validation remains incomplete, and D1 batch-aware persistence is not implemented.

Keywords—multi-agent systems, scholarly search, literature review automation, DOI verification, tool use, reproducibility.

1 Introduction

Literature review is a recurring bottleneck for business-school researchers, graduate students, and project teams. The task is not merely to locate papers whose titles appear related to a question. A defensible review must identify relevant work, verify bibliographic metadata, retain DOI evidence, comply with an approved journal policy, inspect access paths, and explain why specific papers were ranked or excluded. Researchers often perform these checks across disconnected search portals, publisher pages, DOI registries, open-access services, spreadsheets, and local notes.

Generative AI can reduce search effort, but a single LLM call is an inadequate control surface for a scholarly workflow. A fluent answer may contain plausible yet unverifiable recommendations. Even when recommendations are real, the user may not know which external source was consulted, whether journal-policy constraints were applied, whether DOI metadata was checked, or why one result ranked above another. This traceability gap is especially important in academic contexts where the user remains responsible for the final literature selection.

Paper Agent addresses this gap through a traceable multi-agent and tool-using pipeline. Rather than treating an LLM response as the result, the system treats externally verifiable artifacts as the result: candidate metadata, allowlist decisions, Crossref verification outcomes, Unpaywall status, ranking components, persisted traces, and downloadable reports. Its primary user is a researcher preparing an early-stage business-school literature review. Its design goal is reproducible decision support, not autonomous scholarship.

This project makes five contributions:

1. a traceable 12-stage multi-agent architecture for scholarly paper discovery;
2. a DOI- and journal-policy-aware verification pipeline;
3. Paper-Agent-Bench, a domain-specific evaluation protocol;
4. a controlled baseline comparison plus artifact-only staged expansion evidence; and
5. a reproducibility and claim-boundary framework for honest reporting.

1.1 Problem Definition and Design Requirements

The operational problem can be expressed as a constrained evidence-selection task. Given a research question, Paper Agent must retrieve candidate papers, apply a domain-specific venue policy, verify available metadata, score relevance, preserve the origin of each decision, and deliver reviewable outputs. A useful answer is therefore not only a ranked list. It is a ranked list accompanied by evidence that another researcher can inspect.

We derive six design requirements from this problem. First, retrieval and recommendation must be separated so that a candidate can be checked before it is promoted. Second, the venue-selection policy must be explicit. Third, DOI and title metadata must be verifiable through a structured external service. Fourth, intermediate states and failures must be observable. Fifth, generated reports must link back to stored records and traces. Sixth, evaluation claims must remain bounded by the actual scope of audited evidence.

These requirements clarify why a single conversational answer is insufficient. A single LLM response can help generate hypotheses, but it does not automatically expose its retrieval source, enforce an allowlist, preserve failed requests, or make artifact boundaries reproducible. Paper Agent uses agent decomposition to turn these requirements into explicit execution stages.

1.2 Representative User Workflow

A representative workflow begins when a business-school researcher submits a topic such as digital transformation, organizational resilience, or ESG disclosure. The Planner normalizes the request and runtime constraints. The Journal Selector interprets the selected field and prioritizes the approved journal pool. The Retriever submits structured queries to the configured provider. The remaining stages enrich, filter, score, rank, critique, and report the resulting records. The dashboard exposes progress rather than presenting a single opaque completion event.

The user still performs scholarly judgment. A high-ranked result is a candidate for review, not an automatically accepted citation. The system is designed to reduce repetitive verification effort and improve observability while preserving the researcher’s responsibility for relevance, methodology, and interpretation.

2 Related Work

2.1 LLM-Based Scholarly Search

LLM-based assistants can summarize papers, reformulate questions, and generate candidate reading lists. Their usefulness depends on grounding and metadata verification. Large scholarly corpora such as S2ORC show the scale and structure of the evidence that research-support systems must handle [1]. Ungrounded recommendations are

risky in scholarly settings because a fabricated title, DOI, venue, or author list can consume researcher time and contaminate a review. Paper Agent therefore separates recommendation from verification and stores evidence for downstream inspection.

2.2 Multi-Agent and Tool-Using Systems

Multi-agent systems decompose complex tasks into role-specific components such as planners, executors, critics, and reporters. AutoGen demonstrates how multi-agent conversation can structure LLM application workflows [2]. The value of decomposition is not the number of named agents alone; it is the ability to expose intermediate decisions and isolate failures. Paper Agent applies this principle to scholarly discovery by recording stage-level traces and preserving partial outcomes when a later service fails.

2.3 Retrieval-Augmented Workflows

Retrieval-augmented generation grounds model behavior in external evidence [3]. Paper Agent follows the same grounding principle but emphasizes structured scholarly metadata and policy enforcement. Search results are filtered through an internal business-school journal allowlist and enriched through external metadata services before ranking.

2.4 Benchmarking and Reproducibility

Evaluation of agentic systems requires more than a single aggregate score. AgentBench provides an example of evaluating LLMs as agents across multiple environments and highlights the importance of explicit evaluation design [4]. Reproducible evaluation should disclose task scope, baselines, metrics, failure evidence, artifact locations, and incomplete runs. Paper Agent distinguishes controlled benchmark evidence from artifact-only expansion evidence.

3 Method and System Design

Paper Agent is a multi-agent decision-support pipeline rather than a single LLM wrapper. Each stage records an inspectable output. The current Worker implementation uses Web of Science as the primary configured scholarly search provider and OpenAlex as a fallback for testing and quota resilience. Crossref performs DOI and metadata verification, while Unpaywall records open-access status. Cloudflare Workers execute the pipeline and Cloudflare D1 persists jobs, papers, scores, and traces.

3.1 Traceability and Failure Isolation

The pipeline records job progress and stage-level traces in D1. This preserves observable partial state: retrieval failure, timeout, OA lookup failure, or report-generation

Stage	Agent Role	Input	Tool / Data Source	Output Trace
1	Planner	question, constraints	request payload	normalized scope
2	Journal Selector	scope, field	internal allowlist	journal pool
3	Search / Retriever	keywords, years	WoS primary; OpenAlex fallback	candidates
4	Journal Filtering	candidates	internal allowlist	accepted/excluded rows
5	Verifier	DOI metadata	Crossref	verification reasons
6	Open Access	verified DOI	Unpaywall	OA status and URLs
7	Storage	records, artifacts	D1, R2, optional Drive path	persistence trace
8	Evaluation Prep	enriched records	scoring preparation	normalized inputs
9	Relevance	metadata text	metadata scoring; optional semantic path	relevance components
10	Ranking	score components	deterministic ranking	ordered results
11	Critic	ranked results	rule-based default; opt-in LLM path	critic flags
12	Report / Delivery	results, traces	report generator, dashboard	downloadable outputs

Table 1: Twelve-stage Paper Agent workflow. Each stage has an explicit trace boundary.

limitations can be recorded without silently rewriting earlier evidence. The staged benchmark wrappers similarly write isolated artifact directories and stop later batches after a task-level failure.

3.2 Journal Policy and Verification

The internal business-school allowlist is a deliberate constraint. It helps the system focus on journals recognized by the project domain, but it is not a universal quality oracle. The Journal Selector uses the selected business field and tier priorities to construct a search and filtering policy. The filtering stage then records whether a candidate remains eligible. This separation avoids presenting a venue-policy decision as a relevance judgment.

For eligible candidates, the Verifier uses Crossref to inspect DOI and metadata consistency. The pipeline records whether title, year, and journal metadata match sufficiently for downstream review. A verification outcome does not prove substantive quality; it reduces bibliographic uncertainty. The Open Access stage separately uses Unpaywall to record whether an OA PDF or landing page is available. Access status is retained as metadata rather than treated as an academic-quality signal.

3.3 Scoring, Ranking, and Critic Review

Ranking combines observable components such as relevance, journal fit, verification status, citation metadata, recency, and access metadata. The implementation retains metadata-oriented scoring as the inspectable default. Semantic relevance and LLM-backed critic analysis remain optional paths because external model latency and resource constraints can affect live execution. The Critic stage does not overwrite the retrieved record. It adds review flags and summaries so that uncertainty remains visible.

This design supports a conservative interpretation of automation. A ranking score is a prioritization aid, not a scholarly verdict. Researchers can inspect the score components, metadata evidence, and trace history when deciding whether to read, cite, exclude, or re-query a paper.

3.4 Storage, Reporting, and Dashboard Surfaces

Cloudflare D1 stores jobs, papers, evaluations, and stage traces. Cloudflare R2 is used for report-output persistence where configured. The report pipeline exposes CSV, Markdown, XLSX, and PDF outputs, while the dashboard presents research, operations, and evaluation views. The dashboard also distinguishes live D1 evidence, controlled benchmark evidence, legacy partial artifacts, mock blueprint elements, simulation panels, and planned-only features. These labels prevent interface polish from being mistaken for implementation or evaluation evidence.

3.5 Why the Architecture Is Agentic

The system is not a thin interface around one prompt. It combines role decomposition, retrieval tools, explicit verification, persistent traces, deterministic scoring, a critic stage, and report delivery. LLM-backed behavior is optional where appropriate; the default pipeline remains inspectable even when resource-intensive model calls are disabled.

4 Implementation Architecture

4.1 Request Lifecycle

A Paper Agent request starts with a research keyword, optional year boundaries, a result-size limit, and an optional journal-field selection. The Worker creates a search job immediately and exposes a job identifier to the dashboard. This asynchronous boundary is important: the browser does not need to keep a single request open while retrieval and enrichment continue. The dashboard polls for observable progress and can present the latest completed state, while the Worker advances the job through its stage traces.

The Planner records normalized runtime constraints. The Journal Selector resolves the field context and internal allowlist priorities. The Retriever then invokes the config-

ured scholarly backend. In the deployed implementation, Web of Science is the primary provider, and OpenAlex is retained as a fallback for testing and quota resilience. Provider fallback is recorded in traces so that downstream users can distinguish a primary-provider result from a fallback result.

After retrieval, journal filtering applies the approved business-school allowlist. This stage intentionally separates source retrieval count from accepted count. A low accepted count is not necessarily a search failure: it may indicate that the source returned papers outside the policy pool. The Verifier then calls Crossref for DOI-based metadata enrichment. Unpaywall lookup follows as a separate access-status stage. Separating these stages prevents an OA failure from being misreported as a DOI failure or a journal-policy failure.

4.2 Trace and Persistence Model

D1 stores search jobs, candidate papers, evaluations, and agent traces. A trace includes stage order, stage identifier, agent name, status, summary, detail, input count, output count, timestamps, and optional error messages. This representation serves two audiences. Researchers can inspect the human-readable summaries, while operators can use the counts and timestamps to diagnose a stall or external-service failure.

The trace model also supports honest incompleteness. A pipeline does not need to erase a partial result when a later stage fails. For example, a retrieval timeout can be represented as a task-level failure while other tasks in an isolated batch retain their artifacts. This design differs from a product dashboard that displays only a final success badge. Paper Agent treats the path to a result as part of the result.

The benchmark schema has been extended with fields intended for batch orchestration, including parent run identifiers, batch identifiers, derived-run markers, merge status, retry count, and task-level error fields. However, the current staged artifact wrappers do not persist these batch fields to Production D1. This limitation is explicitly retained in the manuscript because schema availability and end-to-end persistence are different implementation claims.

4.3 Evidence and Artifact Layers

The project distinguishes several evidence layers. *Live D1 verified* data describes runtime jobs and traces obtained from deployed services. *Controlled T001–T003* evidence describes the audited quantitative comparison layer. *Artifact-only dry-run* data describes isolated CSV outputs created through gated wrappers. *Legacy partial artifacts* preserve earlier incomplete attempts and failure records. Finally, *mock blueprint* and *scenario simulation* panels support interface design but are not live benchmark evidence.

This taxonomy is enforced in both documentation and dashboard labels. The distinction matters because an interactive interface can look complete before every evaluation path has been audited. By separating evidence sources, the system avoids turning mock values, partial artifacts, or legacy runtime records into final claims.

4.4 Output Surfaces

The report pipeline exposes multiple artifact formats for different uses. CSV and XLSX files support analysis and inspection. Markdown supports readable review summaries. PDF supports distribution, subject to the current report-engine language constraints. Dashboard panels expose ranked papers, paper detail, workflow progress, system diagnostics, and available output links. R2-backed persistence is available for configured report outputs, while optional Drive-related paths remain distinct from the core metadata workflow.

Each output format should be interpreted as a view over stored evidence, not as an independent source of truth. A PDF can summarize a run, but it cannot establish that an unaudited benchmark scope was completed. The report pipeline therefore inherits the same claim boundaries as the underlying jobs, traces, and artifacts.

4.5 Operational Safety Gates

Benchmark expansion uses explicit wrappers rather than broad package commands. Wrappers expose plan and pre-flight modes, isolate artifact paths by run ID, refuse protected directories, reject broad scopes, require explicit acknowledgement flags for execution, and preserve local outputs for read-only inspection. The T007 timeout demonstrates why this gate matters. After Batch 1 produced a failure, the staged process stopped before T013–T018 and T019–T020. A one-shot run could have obscured the point at which resource pressure became significant.

The gate design does not solve every operational issue. It does not implement D1 batch-aware persistence, automatic resumability, or an approved retry policy. Instead, it creates a conservative foundation for later work. The current system favors evidence preservation over aggressive completion.

5 Benchmark Design

Paper-Agent-Bench evaluates whether a literature-review agent retrieves useful and defensible papers under business-school constraints. Tasks represent domain questions and keyword sets. Gold labels are DOI-backed paper records reviewed under a relevance rubric. The controlled quantitative layer currently covers T001–T003. Expansion runs beyond this layer are reported separately as artifact-only evidence until evaluation and audit are complete.

Metric	Definition and purpose
Precision@5	Fraction of top-five results judged relevant; measures immediate usefulness.
NDCG@5	Rank-sensitive normalized gain; rewards ordering of highly relevant results.
DOI Accuracy	Fraction with correct DOI evidence; measures metadata integrity.
Top Journal Precision	Fraction satisfying the approved journal policy.
Hallucination Rate	Fraction of unsupported recommendations; measures trust risk.

Table 2: Paper-Agent-Bench metrics.

5.1 Baselines

We compare three approaches: (1) a *Rule-Based* baseline with deterministic retrieval and ranking heuristics, (2) a *Single LLM* recommendation baseline, and (3) *Paper Agent*, the traceable multi-agent pipeline.

5.2 Task and Gold-Label Construction

Paper-Agent-Bench is organized around business-research topics rather than generic question answering. Each task contains a keyword query and a journal-category context. Gold-label construction requires DOI-backed scholarly papers and records enough metadata for follow-up verification. Ambiguous candidates must remain ambiguous rather than being silently promoted. This matters because an incorrect gold set can reward confident retrieval while penalizing defensible abstention.

The controlled T001–T003 layer is intentionally narrow. It establishes an auditable starting point with nine comparison rows across the three methods. The remaining tasks are expansion targets, not automatically comparable observations. They require staged execution, artifact inspection, and subsequent evaluation before they can support metric claims.

5.3 Metric Interpretation

Precision@5 and NDCG@5 measure overlap and ordering relative to the gold labels. DOI Accuracy and Top Journal Precision measure integrity and policy compliance. Hallucination Rate captures unsupported recommendation risk. None of these metrics should be interpreted in isolation. For example, a method can achieve high overlap while offering weak traceability, or strict policy filtering can reduce overlap while increasing compliance. We therefore report a multi-dimensional evaluation rather than a single winner score.

5.4 Protocol and Reproducibility Controls

Quantitative claims are limited to the controlled T001–T003 layer with nine comparison rows. Execution artifacts

from T004–T006 and T007–T012 are isolated by run ID and are not merged into controlled metrics. Gold labels are not rewritten to hide failures. The repository retains scripts, prompts, trace APIs, documentation reports, and claim-boundary rules. Live reproduction requires valid API credentials and available external services.

6 Experiments

6.1 Controlled T001–T003 Comparison

The verified controlled layer compares Rule-Based, Single LLM, and Paper Agent baselines on T001–T003. It contains nine comparison rows. This is the only layer used for quantitative baseline claims in this manuscript.

6.2 T004–T006 Artifact-Only Dry-Run

An explicitly approved gated wrapper executed T004–T006 as an artifact-only dry-run. T004 generated 20 rows, T005 generated 12 rows, and T006 generated 18 rows. The run produced three job rows and 50 result rows without failed jobs. These artifacts demonstrate execution coverage, not benchmark validation.

6.3 Phase 3L Partial Staged Expansion

Phase 3L produced partial staged expansion evidence. Batch 1 (T007–T012) generated artifacts for T008–T012, while T007 timed out after approximately 250 seconds. Because this failure occurred within Batch 1, later batches T013–T018 and T019–T020 were not started. This result identifies both executable segments and infrastructure boundaries, and it does not constitute full benchmark validation.

6.4 Artifact Isolation Strategy

The staged execution wrappers enforce isolated output directories named by run ID. This prevents new attempts from overwriting controlled artifacts or earlier dry-run evidence. Before execution, wrappers expose plan and pre-flight modes. Execute mode requires explicit acknowledgement flags and an exact approved run ID. No permanent execute command is exposed through the package scripts. This operational design makes scope expansion reviewable and limits accidental broad execution.

6.5 What Was Not Evaluated

We do not report a completed T004–T020 benchmark. Artifact-only evidence is not merged into controlled metrics. D1 batch-aware persistence is not implemented. We do not claim global superiority over the Single LLM baseline.

Metric	Rule	Single LLM	Paper Agent
Precision@5	0.1333	0.6667	0.1333
NDCG@5	0.3579	0.9949	0.3579
DOI Accuracy	1.0000	1.0000	1.0000
Top Journal Prec.	1.0000	0.9333	1.0000
Hallucination Rate	0.0	risk noted	0.0

Table 3: Controlled T001–T003 comparison only.

Task	Rows	Notes
T004	20	completed
T005	12	completed
T006	18	completed
Total	50	artifact-only evidence

Table 4: T004–T006 gated artifact-only dry-run.

7 Results and Discussion

The Single LLM baseline may appear strong under gold-label overlap metrics, but its recommendation process remains less traceable and less constrained by the explicit journal-policy and metadata-verification pipeline. Paper Agent does not outperform the Single LLM baseline on Precision@5 or NDCG@5 in the controlled scope. Its current contribution is a different tradeoff: DOI integrity, journal-policy compliance, traceability, and preserved failure evidence. Treating overlap metrics as the only objective can create an interpretive trap, but this is a risk to examine rather than a proven causal explanation.

Batch 2 T013–T018 and Batch 3 T019–T020 were not started after the T007 timeout. Legacy T019–T020 resource-limit / HTTP 503 evidence remains visible. The safety stop is a meaningful system result: execution coverage should expand only when failure evidence remains inspectable.

7.1 Discussion of the Controlled Tradeoff

The controlled table should not be compressed into a winner narrative. The Single LLM baseline performs better on overlap-oriented metrics in T001–T003. Paper Agent matches the Rule-Based baseline on those metrics in the current controlled scope. At the same time, Paper Agent exposes an operationally different evidence surface: journal-policy filtering, structured metadata verification, stored traces, separated OA status, report artifacts, and failure preservation.

This tradeoff is relevant for literature-review support because the cost of a recommendation is not limited to ranking error. Researchers also incur costs when they must investigate fabricated metadata, reconstruct an opaque decision, or discover that a polished interface mixed simulated and live evidence. The current experiment does not quantify all of these costs. It identifies them as dimensions for a broader benchmark.

Task	Rows	Notes
T007	0	timeout: 250250ms / 21 attempts
T008	20	completed
T009	20	completed
T010	20	completed
T011	7	completed
T012	20	completed
Total	87	partial artifact evidence

Table 5: Phase 3L Batch 1 partial staged expansion.

7.2 Operational Failure Analysis

The T007 timeout provides a concrete operational boundary. The runner waited approximately 250 seconds across 21 polling attempts before recording a timeout. It then preserved the failure and continued through the remaining tasks in Batch 1. The outer staged process stopped before Batch 2 and Batch 3. This behavior demonstrates containment: the system retained successful T008–T012 artifacts without hiding T007 or escalating into an uncontrolled broad run.

Legacy T019–T020 HTTP 503 records provide a second type of evidence. They indicate that external-service or Worker resource constraints must remain part of the evaluation story. The proper response is not to erase failed rows or relabel them as success. Future work should improve timeout handling, resource budgeting, and retry policies while keeping failure visibility intact.

7.3 Threats to Validity

There are at least four threats to validity. First, the controlled task set is small, so performance metrics cannot establish broad generalization. Second, gold-label quality depends on manual verification and may underrepresent relevant papers outside the initial pool. Third, live API behavior is time-dependent: quotas, provider responses, and network conditions can change. Fourth, the internal journal allowlist encodes institutional preferences that may not transfer to every research context. These threats motivate staged evidence collection and conservative claims.

8 Limitations and Ethics

The current evidence has clear boundaries. Full T004–T020 validation remains incomplete. D1 batch-aware persistence is not implemented. Phase 3L recorded a T007 timeout after approximately 250 seconds, and T013–T020 were not executed under Phase 3L after the safety stop. Gold-label coverage remains limited and should be expanded before broader performance claims.

Paper Agent also introduces normative choices. The internal journal allowlist can create algorithmic gatekeeping by prioritizing established venues and potentially excluding useful work from emerging, regional, or interdisciplinary.

plinary outlets. Automated ranking can amplify citation and journal-hierarchy biases. DOI verification improves metadata integrity but does not establish substantive research quality. Open-access status can improve accessibility but should not be interpreted as a quality signal.

Human scholarly review remains the final authority. Users must inspect recommendations, verify metadata when necessary, evaluate methodological fit, and take responsibility for citation choices. Responsibility for hallucination, metadata error, and ranking bias cannot be delegated entirely to an automated pipeline.

9 Conclusion

Paper Agent is a traceable multi-agent pipeline for business-school scholarly discovery. It combines role decomposition, retrieval tools, DOI verification, journal-policy filtering, open-access checking, persistent traces, ranking, critic review, and report delivery. The verified T001–T003 controlled comparison shows a nuanced result: the Single LLM baseline has higher overlap-based metrics, while Paper Agent preserves strict top-journal precision, DOI integrity, traceability, and hallucination control within the controlled scope.

Artifact-only expansion provides additional implementation evidence without changing the benchmark claim boundary. T004–T006 generated 50 result rows through a gated wrapper. Phase 3L generated 87 result rows for T008–T012 while preserving the T007 timeout and stopping later batches. Full T004–T020 validation remains incomplete.

10 Reproducibility Statement

10.1 Repository Evidence

The GitHub repository contains source code, benchmark scripts, prompts, documentation reports, staged-wrapper logic, and a paper claim-boundary checklist. Cloudflare D1 stores live jobs and traces, while isolated local artifact directories preserve staged-run outputs. The controlled benchmark and expansion artifacts are intentionally described as separate evidence layers.

10.2 Reproduction Procedure

A reproduction attempt should first inspect repository status, configuration, and documented safety boundaries. Live execution requires separately provisioned API credentials and available external services. Benchmark expansion must use plan and preflight modes before any approved execute mode. Artifact directories must be isolated by run ID, and controlled T001–T003 evidence must remain unchanged. After execution, researchers should inspect job CSV files, result CSV files, checksums, status rows, failure messages, and trace records before making claims.

10.3 Reproduction Constraints

Reproduction depends on API credentials, external service availability, and explicit approval for gated execution. The current codebase does not implement D1 batch-aware persistence for staged artifacts. Consequently, local artifact inspection and documentation remain necessary. Future work includes separately approved T013–T020 staged expansion, D1 batch-aware persistence, broader gold-label coverage, improved timeout handling, and optional external bibliometric enrichment where appropriate.

11 Future Work

The next engineering priority is not a broader claim. It is a safer evidence pipeline. First, the team should diagnose the T007 timeout using the preserved polling evidence and determine whether Worker CPU limits, external API latency, enrichment volume, or report generation caused the delay. Any retry policy should be bounded and documented. Second, the batch-aware D1 fields should be connected to the staged wrappers so that parent run ID, batch ID, retry count, last error, and last-error timestamp are persisted end to end.

Third, Paper-Agent-Bench should expand its DOI-backed gold labels and record clearer provenance for ambiguous rows. A broader controlled layer would allow the team to examine whether journal-policy filtering trades overlap for integrity in a consistent way across fields. Fourth, evaluation should include operational metrics such as latency, resource-limit incidence, trace completeness, and recovery behavior. These dimensions capture costs that Precision@5 and NDCG@5 alone cannot describe.

Fifth, the journal-policy layer should support explicit alternatives. Researchers may need a strict internal allowlist, a broader discovery mode, or an externally enriched bibliometric view. Such modes should be visible and should not silently change the meaning of a score.

12 Implementation and Evaluation Checklist

Before final submission, the team should audit the T001–T003 metric provenance, inspect untracked staged artifacts, and retain the claim boundary. Any later T013–T020 execution requires a separate explicit approval. A completed dashboard or generated report must not be used as evidence of a completed benchmark unless the underlying artifacts and evaluation outputs have been audited.

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